WO-975
Job
STOCK
Customer

FORM U-1 MANUFACTURER'S DATA REPORT FOR PRESSURE VESSELS

As Required by the Provisions of the ASME Boiler and Pressure Vessel Code Rules, Section VIII, Division 1

1. Manufactured and certified by **PHOENIX INNOVATIONS, INC., 4254 South Arkansas Ave., Russellville, Arkansas, 72802, USA**
(Name and address of Manufacturer)

2. Manufactured for **STOCK**
(Name and address of Purchaser)

3. Location of installation **UNKNOWN**
(Name and address)

4. Type **Horizontal** (Horizontal, vertical, or sphere) | **HEAT EXH** (Tank, separator, jkt. vessel, heat exch., etc.) | **PPV-145** (Manufacturer's serial number)

- (CRN) | **504952 REV 0** (Drawing number) | **123** (National Board number) | **2025** (Year built)

5. ASME Code, Section VIII, Div. 1 **2023/ NO AD** [Edition and Addenda, if applicable (date)] | **-** (Code Case Number) | **-** [Special Service per UG-120(d)]

Items 6-11 incl. to be completed for single wall vessels, jackets of jacketed vessels, shell of heat exchangers, or chamber of multichamber vessels.

6. Shell: (a) Number of course(s) **1** (b) Overall length **19' 7"**

Course(s)			Material	Thickness		Long. Joint (Cat. A)			Circum. Joint (Cat. A, B, & C)			Heat Treatment	
No.	Diameter	Length	Spec./Grade or Type	Nom.	Corr.	Type	Full, Spot, None	Eff.	Type	Full, Spot, None	Eff.	Temp.	Time
1	18" OD	235"	SA-53 B ERW	.375	0.00	1	NONE	.70	1	NONE	.70	-°	-

Body Flanges on Shells

No.	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Location	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
-	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

7. Heads: (a) **N/A** (Material spec. number, grade or type) (H.T. - time and temp.) (b) **N/A** (Material spec. number, grade or type) (H.T. - time and temp.)

	Location (Top, Bottom, Ends)	Thickness		Radius		Elliptical Ratio	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure		Category A		
		Min.	Corr.	Crown	Knuckle					Convex	Concave	Type	Full, Spot, None	Eff.
(a)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			N/A	N/A	N/A
(b)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			N/A	N/A	N/A

Body Flanges on Heads

No.	Location	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
(a)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		N/A	N/A	N/A

8. Type of jacket **N/A** Jacket closure **N/A** (Describe as ogee & weld, bar, etc.)

If bar, give dimensions; if bolted, describe or sketch **N/A**

9. MAWP **250 psi** (Internal) **N/A** (External) at max. temp. **350 °F** (Internal) **N/A** (External) Min. design metal temp. **-20 °F** at **250 psi**

10. Impact test **NO** [Indicate yes or no and the component(s) impact tested] at test temperature of **-**

11. Hydro., pneu., or comb. test pressure **Hydro. at 325 psi** Proof test **-**

Items 12 and 13 to be completed for tube sections.

12. Tubesheet **SA 240 TP 304** [Stationary (material spec. no.)] | **18.00"** [Diameter (subject to press.)] | **1.0** (Nominal thickness) | **0.00** (Corr. allow.) | **WELDED** Attachment (welded or bolted)

N/A [Floating (material spec. no.)] | **N/A** (Diameter) | **N/A** (Nominal thickness) | **N/A** (Corr. allow.) | **N/A** (Attachment)

13. Tubes **SA 312 304EFW** (Material spec. no., grade or type) | **2.5** (O. D.) | **.12** (Nominal thickness) | **8** (Number) | **Straight** [Type (Straight or U)]

FORM U1

Manufactured by **PHOENIX INNOVATIONS, INC., 4254 South Arkansas Ave., Russellville, Arkansas, 72802, USA**

Manufacturer's Serial No. **PPV-145** CRN **-** National Board No. **123**

Items 14-18 incl. to be completed for inner chambers of jacketed vessels or channels of heat exchangers.

14. Shell: (a) No. of course(s) **N/A** (b) Overall length **N/A**

Course(s)			Material	Thickness		Long. Joint (Cat. A)			Circum. Joint (Cat. A, B, & C)			Heat Treatment	
No.	Diameter	Length	Spec./Grade or Type	Nom.	Corr.	Type	Full, Spot, None	Eff.	Type	Full, Spot, None	Eff.	Temp.	Time
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**

Body Flanges on Shells

No.	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Location	Bolting: Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**

15. Heads: (a) **N/A** (Material spec. number, grade or type) (H.T. - time and temp.) (b) **N/A** (Material spec. number, grade or type) (H.T. - time and temp.)

	Location (Top, Bottom, Ends)	Thickness		Radius		Elliptical Ratio	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure		Category A		
		Min.	Corr.	Crown	Knuckle					Convex	Concave	Type	Full, Spot, None	Eff.
N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**

Body Flanges on Heads

No.	Location	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Bolting: Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**

16. MAWP **N/A** (Internal) **N/A** (External) at max. temp. **N/A** (Internal) **N/A** (External) Min. design metal temp. **N/A** at **N/A** .

17. Impact test **N/A** [Indicate yes or no and the component(s) impact tested] at test temperature of **N/A** .

18. Hydro., pneu., or comb. test pressure **Hydro. at 325 psi** Proof test **N/A**

19. Nozzles, inspection, and safety valve openings:

Purpose (Inlet, Outlet, Drain, etc.)	No.	Diameter or Size	Type	Material		Nozzle Thickness		Reinforcement Material	Attachment Details		Location (Insp. Open.)
				Nozzle	Flange	Nom.	Corr.		Nozzle	Flange	
LIQUID INLET	**1**	**1"**	**NONE**	**SA 106B**	**-**	**S/80**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**SHELL**
COLUMN	**2**	**1 1/4"**	**NONE**	**SA 106B**	**-**	**S/80**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**HEADER/SUMP**
HEADER/SHELL	**4**	**3"**	**NONE**	**SA 106B**	**-**	**S/40**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**HEADER&SHELL**
SUMP	**1**	**8"**	**NONE**	**SA 106B**	**-**	**S/40**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**SHELL**
DRAIN	**1**	**1"**	**FPT**	**SA 105**	**-**	**3M**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**SUMP CAP**
SUCTION	**1**	**2"**	**NONE**	**SA 106B**	**-**	**S/80**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**HEADER**

Additional Nozzles - See Attached U-4...

20. Supports: Skirt **No** (Yes or no) Lugs **2**** (Number) Legs **2 SQ TUBE LEGS ***** (Number) Others **N/A** (Describe) Attached **WELDED TO SHELL WELD PAD** (Where and how)

21. Manufacturer's Partial Data Reports properly identified and signed by Commissioned Inspectors have been furnished for the following items of the report (list the name of part, item number, Manufacturer's name, and identifying number):

N/A

22. Remarks

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Length of tubes: 240.0"
** 1/2" SA 36 LUGS WELDED TO SHELL
*** WELDED TO WELD PADS ON SHELL BOTTOM 1/2" SA 36 LEGS ARE 3"X3"X1/4THK 304SS 'T' STYLE
8" S/40 NOMINAL HEADER SA-53BERW 189.25 INCHES LONG WITH STD SA 234 WPB BW CAPS ATTACHED TO SHELL
VIA 3" NOZZLES
QTY 2:  1/2" THICK SA 36 WELD PADS PROVIDED FOR PIPE/VALVE SUPPORT ON SHELL AND HEADER.
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Form U-5 Attached....

FORM U1

Manufactured by **PHOENIX INNOVATIONS, INC., 4254 South Arkansas Ave., Russellville, Arkansas, 72802, USA**

Manufacturer's Serial No. **PPV-145** CRN **-** National Board No. **123**

CERTIFICATE OF SHOP COMPLIANCE

We certify that the statements in this report are correct and that all details of design, material, construction, and workmanship of this vessel conform to the ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. U Certificate of Authorization Number **59994** Expires **February 18, 2028**

Date 07/22/2025 Name **PHOENIX INNOVATIONS, INC.** (Manufacturer) Signed ____ (Representative)

CERTIFICATE OF SHOP INSPECTION

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and employed by **Bureau Veritas Inspection and Insurance Company, of Lynn, MA** have inspected the pressure vessel described in this Manufacturer's Data Report on **July 22, 2025**, and state that, to the best of my knowledge and belief, the Manufacturer has constructed this pressure vessel in accordance with ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. By signing this certificate neither the Inspector nor his/her employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturer's Data Report. Furthermore, neither the Inspector nor his/her employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date 07/22/2025 Signed ____ (Authorized Inspector) Commissions: **9731, AR1180, AK20250046 LA1394, MS3563, TX1453, KS786, OK2338, TN4289** (National Board Authorized Inspector Commission number)

CERTIFICATE OF FIELD ASSEMBLY COMPLIANCE

We certify that the statements made in this report are correct and that the field assembly construction of all parts of this vessel conforms with the requirements of ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. U Certificate of Authorization Number ____ Expires ____

Date ____ Name ____ (Assembler) Signed ____ (Representative)

CERTIFICATE OF FIELD ASSEMBLY INSPECTION

I, the undersigned, holding a valid commission issued by The National Board of Boiler and Pressure Vessel Inspectors and employed by ____, have compared the statements in this Manufacturer's Data Report with the described pressure vessel and state that parts referred to as data items ____, not included in the certificate of shop inspection, have been inspected by me and to the best of my knowledge and belief, the Manufacturer has constructed and assembled this pressure vessel in accordance with the ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. The described vessel was inspected and subjected to a pressure test of ____ . By signing this certificate neither the Inspector nor his/her employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturer's Data Report. Furthermore, neither the Inspector nor his/her employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date ____ Signed ____ (Authorized Inspector) Commission ____ (National Board Authorized Inspector Commission number)

FORM U-4 MANUFACTURER'S DATA REPORT SUPPLEMENTARY SHEET

As Required by the Provisions of the ASME Boiler and Pressure Vessel Code Rules, Section VIII, Division 1

1. Manufactured and certified by **PHOENIX INNOVATIONS, INC., 4254 South Arkansas Ave., Russellville, Arkansas, 72802, USA**
(Name and address of Manufacturer)

2. Manufactured for **STOCK**
(Name and address of Purchaser)

3. Location of installation **UNKNOWN**
(Name and address)

4. Type **Horizontal** (Horizontal, vertical, or sphere) **HEAT EXH** (Tank, separator, heat exch., etc.) **PPV-145** (Manufacturer's serial number)

- (CRN) **504952 REV 0** (Drawing number) **123** (National Board number) **2025** (Year built)

Data Report Item Number	Remarks

Additional nozzles, inspection and safety valve openings:

Purpose (Inlet, Outlet, Drain, etc.)	No.	Diameter or Size	Type	Material		Nozzle Thickness		Reinforcement Material	Attachment Details		Location (Insp. Open.)
				Nozzle	Flange	Nom.	Corr.		Nozzle	Flange	
RELIEF	**1**	**1 1/2"**	**FPT**	**SA 105**	**-**	**3M**	**0.00**	**INHERENT**	**UW-16.1(c)**	**-**	**HEADER**

Certificate of Authorization: Type **"U"** No. **59994** Expires **February 18, 2028**

Date **07/22/2025** Name **PHOENIX INNOVATIONS, INC.** (Manufacturer) Signed (Representative)

Date **07/22/2025** Signed (Authorized Inspector) Commissions 9731, AR1180, AK20250046 LA1394, MS3563, TX1453, KS786, OK2338, TN4289
(National Board Authorized Inspector Commission number)

FORM U-5 MANUFACTURER'S DATA REPORT SUPPLEMENTARY SHEET

SHELL-AND-TUBE HEAT EXCHANGERS

As Required by the Provisions of the ASME Boiler and Pressure Vessel Code Rules, Section VIII, Division 1

1. Manufactured and certified by **PHOENIX INNOVATIONS, INC., 4254 South Arkansas Ave., Russellville, Arkansas, 72802, USA**
(Name and address of Manufacturer)

2. Manufactured for **STOCK**
(Name and address of Purchaser)

3. Location of Installation **UNKNOWN**
(Name and address)

4. Type **Horizontal** (Horizontal, vertical, or sphere) **PPV-145** (Manufacturer's serial number) **-** (CRN)

504952 REV 0 (drawing no.) **123** (National Board number) **2025** (Year built)

FIXED TUBESHEET HEAT EXCHANGERS										
Name of Condition	Design/Operating Pressure Ranges				Design/Operating Metal Temperature				Allowable Axial Differential Thermal Expansion Range	
	Shell Side		Tube Side		Shell	Channel	Tubes	Tubesheet		
	Min.	Max.	Min.	Max.					Min.	Max.
	units:	units:	units:	units:	units:	units:	units:	units:	units:	units:
Design	**0**	**250**	**N/A**	**N/A**	**350**	**N/A**	**N/A**	**350**	**0.00**	**0.00**
OPERATING	**0**	**250**	**N/A**	**N/A**	**20**	**N/A**	**70**	**35**	**0.004**	**0.0725**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**
	N/A	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**

Data Report Item Number	Remarks

VESSEL IS DESIGNED FOR NON-CORROSIVE NON-LETHAL SERVICE
IMPACT EXEMPT PER UHA-51 & UCS-66
INSPECTION OPENINGS OMITTED PER UG-46(b)(4)
HYDRO IN HORIZONTAL POSITION.
*DRAFT*DRAFT*DRAFT*DRAFT*DRAFT*REVIEW INPUTS*

Certificate of Authorization: Type **"U"** No. **59994** Expires **February 18, 2028**

Date **07/22/2025** Name **PHOENIX INNOVATIONS, INC.** (Manufacturer) Signed ______ (Representative)

Date **07/22/2025** Signed ______ (Authorized Inspector) Commissions: 9731, AR1180, AK20250046 LA1394, MS3563, TX1453, KS786, OK2338, TN4289 (National Board Authorized Inspector Commission number)